

$$\hat{X} = BFx; \quad (x)_{D \times N} = \begin{pmatrix} x_1 & \dots & x_N \end{pmatrix}, \quad (F)_{K \times D}, \quad (B)_{D \times K}$$

$$\|x - \hat{x}\|_F^2 = \sum_{i=1}^D (x_{i0} - \hat{x}_{i0})^2 = \text{Tr}((x - \hat{x})(x - \hat{x})^T); \quad \|A\|_F^2 = \text{Tr}(AA^T)$$

SVP of X : $X = U \Sigma V^T$

$$(U)_{D \times D} = \begin{pmatrix} u_1 & \dots & u_D \end{pmatrix}$$

$$(\Sigma)_{D \times N} = \begin{pmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \sigma_N \\ 0 & \dots & 0 & 0 \end{pmatrix} \quad (V)_{N \times N} = \begin{pmatrix} v_1 & \dots & v_N \end{pmatrix}$$

U is orthogonal: $U^T U = U U^T = I$

V is orthogonal too.

$$\|U A V^T\|_F^2 = \text{Tr}[U A V^T V A^T U^T] = \text{Tr}[A A^T \underbrace{V V^T}_I] = \text{Tr}[A A^T] = \|A\|_F^2 \quad \checkmark$$

$$\begin{aligned} \|x - \hat{x}\|^2 &= \|U \Sigma V^T - \hat{x}\|^2 = \|U(\Sigma - \underbrace{U^T \hat{x} (V^T)^{-1}}_{A \text{ above}}) V^T\| \\ &= \|U(\Sigma - U^T \hat{x} V) V^T\| \\ &= \|\Sigma - \underbrace{U^T \hat{x} V}_S\| \end{aligned}$$

$$\text{rank}(S) = k = \text{rank}(\hat{x}) \quad , \quad k \leq D$$

$$\min_{\hat{x}} \|x - \hat{x}\|^2 = \min_{\substack{S \\ \text{rank}(S) = k}} \|\Sigma - S\|$$

Hence:

$$S = \text{Diag}(\sigma_1, \dots, \sigma_k, 0, 0, \dots, 0) \quad \sigma_1 > \sigma_2 > \dots > \sigma_k$$

$D \times N$

$$S = U^T \hat{X} V \Rightarrow \hat{X} = \underbrace{U}_{D \times D} \underbrace{S}_{D \times N} \underbrace{V^T}_{N \times N}$$

$$X = U \bar{S} V^T \Rightarrow \bar{S} = U^T X V$$

$$S = I_k \quad \bar{S} = \quad , \quad (I_k)_{D \times D} = \text{diag} \left(\underbrace{1, \dots, 1}_k, 0, 0 \right)'$$

copies

$$S = I_k \quad U^T X V$$

$$\hat{X} = \underbrace{U}_{D \times D} \underbrace{I_k}_{D \times D} \underbrace{U^T X V V^T}_{I} = \underbrace{U_k}_{W^T} \underbrace{U_k^T}_W X$$

$$\text{where } U_k = (u_1 | u_2 \dots | u_k | 0 | \dots 0)$$

In general, W is not unique !

$$\tilde{W} = U_k^T R \quad ; \quad |R| = 1, \quad R R^T = I, \quad (R)_{k \times k}$$

The content in this document is not examinable and provided only as supplementary material.